

Data Management (A.A. 2025/26) – exam of 7/07/2026

Problem 1. A *return point* in a schedule S over a set of transactions $\{T_1, \dots, T_n\}$ is a sequence of two consecutive actions a_i, b_j in S such that: $i \neq j$ (the actions belong to different transactions). Transaction T_j has already executed at least one action in S prior to b_j (i.e., the schedule "returns" to a previously interrupted transaction). A schedule is said to be *single-return* if it contains exactly one return point. By referring to the transactions: $T_1 = w_1(x)r_1(y)$, $T_2 = w_2(x)r_2(z)$, and $T_3 = w_3(x)w_3(y)$, prove or disprove the following statements, motivating each of your answers in detail.

- 1.1 There is no single-return schedule on transactions $\{T_1, T_2, T_3\}$ that is view-serializable.
- 1.2 There is no single-return schedule on transactions $\{T_1, T_2, T_3\}$ that is conflict-serializable.
- 1.3 There is no single-return schedule on transactions $\{T_1, T_2, T_3\}$ that is rigorous.
- 1.4 There is no single-return schedule on transactions $\{T_1, T_2, T_3\}$ that is strict.

Problem 2. Given the schedule $S = r_1(x), w_3(x), w_2(x), r_3(y), w_1(y)$,

- 2.1 Determine whether S is a valid 2PL (Two-Phase Locking) schedule using both shared and exclusive locks. Motivate your answer.
- 2.2 Assuming that (i) the system clock is 0, (ii) $rts(x) = rts(y) = wts(x) = wts(y) = wts-c(x) = wts-c(y) = 0$, $cb(x) = cb(y) = true$, and (iii) the system uses the clock values as the timestamps to be assigned to the various transactions (at their first action), trace exactly what a Timestamp-Based Scheduler would do when presented with the schedule S .

Problem 3. Let `Student(StudId, StudName)` be a relation stored in a file with 200 pages sorted on `<StudId, StudName>`. Let `Attend(StudId, Course)` be a relation stored in a file with 500 pages sorted on `<StudId, Course>`. Let `Exam(StudId, StudName, Course)` be a relation stored in a file with 400 pages sorted on `<StudId, StudName, Course>` (primary keys are underlined). Assume that exactly 205 buffer frames are available. We want to compute `(Student ⋈ Attend) – Exam`, that is, the natural join between `Student` and `Attend` and then the difference between the join result and `Exam`. Describe the I/O-optimal algorithm for evaluating the expression, justify your choice, and compute its cost in page accesses. Can the same cost be achieved with fewer buffer frames? If so, determine the minimum number required.

Problem 4. Let `Event(EventName, EventCategory, Year, City)` be a relation stored in a file with 50.000 pages. A clustering B⁺-tree strongly dense index, using Alternative 2 for data entries, is available on search key `<City, Year>`. Each page has size 400 KB and each attribute value and each pointer occupy 20 KB. Consider the query

SELECT City, COUNT(*) FROM Event GROUP BY City

Tell which I/O-optimal algorithm you would use to evaluate the query, justify your choice, and compute its cost in terms of page accesses. How would your answer change if the search key of the clustering B⁺-tree strongly dense index were `<Year, City>` instead of `<City, Year>`?

Problem 5. (only for students who opted for **option 1**, i.e., who do **not** do the project)

A car-sharing company manages cars in several cities worldwide. Each car has a license plate, a registration year, and a fuel type. A car belongs to a single city and can only be rented there. Each rental is characterized by a date, a starting time, an ending time, and the customer who rented the car. Each customer has a city of residence and an age. Based on the operational domain described above, the company wants to construct a DW to analyze the number of car rentals and the total rental time (in hours) by date, month, quarter, year, decade, day of the week (Monday, Tuesday, etc.), also distinguishing holidays from other days. Moreover, it wants to analyze car rentals by the city, region, and country in which the rental occurred, and by the city, region and country of residence of the customers who rented the cars. Finally, it wants to maintain in the DW the number of inhabitants of each city, and to analyze rentals by year and decade of registration of the car, as well as by fuel type.

- 5.1 Provide a DFM diagram supporting the above analyses through OLAP operations.
- 5.2 Design the star/snowflake schema corresponding to the DFM diagram.
- 5.3 Suppose that the vast majority of the cars belong to cities located in only a few regions. Would you modify the schema designed above? If so, describe the modification and explain its advantages.
- 5.4 Based on the schema you designed, write the SQL query that returns the number of car rentals for each region of Italy, for each month, and for each fuel type.